ITIS 4250

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Lab 3 - Windows Examination

4/6/2026

Overview

On 4/6/2026, Detective Angela Lansbury and Agent Sampson provided forensic images of two computers for examination under a warrant signed by a North Carolina Superior Court judge. The evidence includes a laptop belonging to Malcom Monarch which were seized from Conjectural Technologies after technician Pete White flagged suspicious activity and images from a Clarion Hotel Internet kiosk believed to have been used by the subject while on the run.

Agent Sampson requested a forensic analysis to find communications, web searches, and media related to the criminal activity of supervillain Malcom Monarch. This includes identifying Guild operatives, email communications, evidence of searches related to explosions or covering tracks, downloaded photos, and research into forensics software.

Forensic Acquisition & Exam Preparation

The examination was performed on my personal laptop - an ASUS Predator Helios running Windows 11 with 16GB of RAM. I created a case folder at C:\Users\sampa\Documents\Monarch_Lab3\ with subfolders for Evidence, Registry, Screenshots, and Exports.

Tools used for this examination:
- FTK Imager 4.7.1
- Autopsy 4.22.1
- Registry Explorer v2.1.0

I verified all three forensic images using FTK Imager. All images passed verification with matching MD5 and SHA1 hashes.

MonarchIMG2021.E01 Verification:
MD5: dd2ddf067505a2a35264528dc43dc448
SHA1: 9d1553b3a4d5e13002a659d0c9f32545fbc3b136
Result: Match

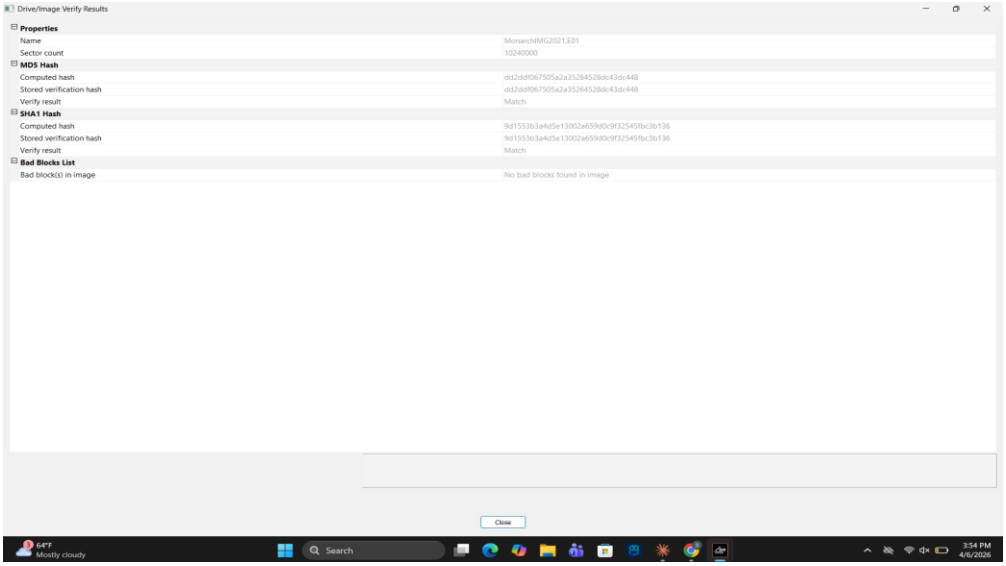

Figure 1: MonarchIMG2021.E01 Hash Verification

OldClarionIMG.E01 Verification:
MD5: 2b07ed2d5e2045aada0d1cda0cb43e5e
SHA1: 026913f9527667cef51eb37ad81da01dcb3a84ce
Result: Match

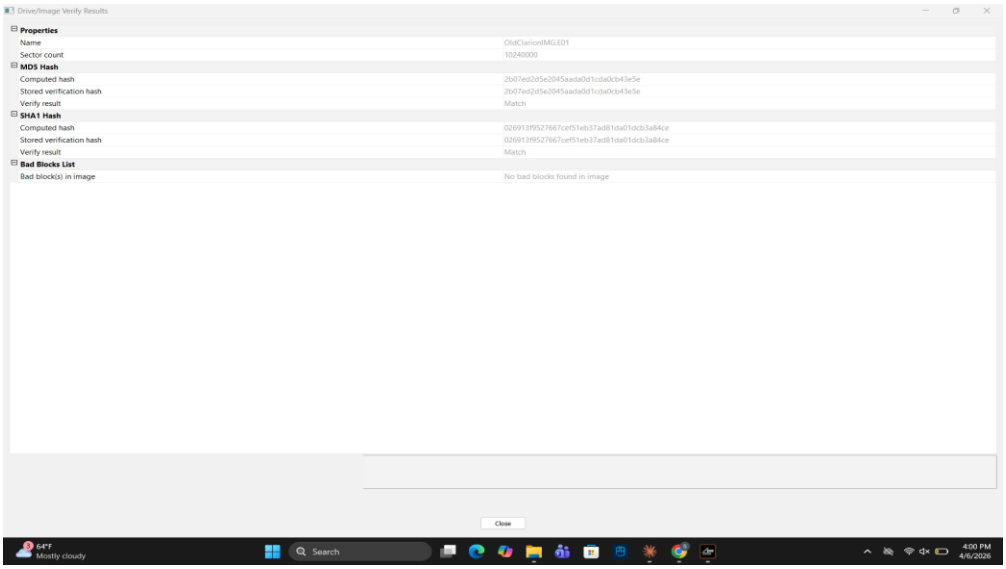

Figure 2: OldClarionIMG.E01 Hash Verification

ClarionBusinessCenter.E01 Verification:
MD5: 81c5fe2491a0bfdd66aaf3c64d4886a1
SHA1: e8910334f911d952504ac15a0e5e612ae2f4ed0d
Result: Match

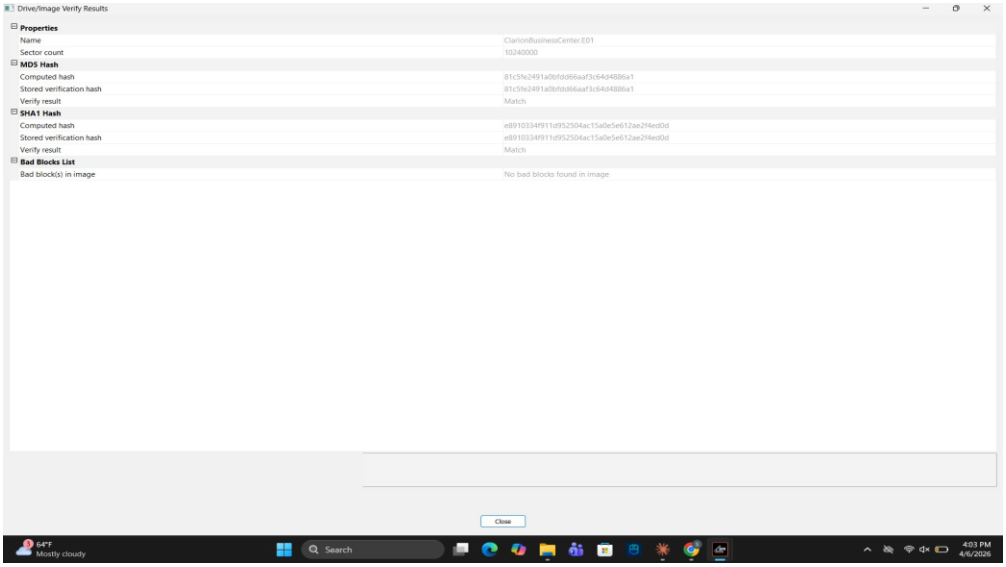

Figure 3: ClarionBusinessCenter.E01 Hash Verification

After verification, I loaded all three images into Autopsy and ran the ingest modules including Recent Activity, Email Parser, Keyword Search, and PhotoRec Carver. I also extracted the registry files from RegistryM.zip and RegistryC.zip for analysis in Registry Explorer.

Findings and Report (Forensic Analysis)

1) Was there communication from someone claiming to be the Guild's operative, or can you identify the operative's name/email?

Yes, I found communication from a Guild operative. The operative is **Hamilton J Fantomos, also known as The Phantom Limb**. His email address is **generalconsolidatedindustries@aol.com**.

I found this by going to Data Artifacts > E-Mail Messages in Autopsy and looking through the emails. The email with subject "Your Application" from generalconsolidatedindustries@aol.com states "The new governance body operating the Guild of Calamitous Intent, by order of the Sovereign and his Council of 13, has received your application..." and is signed by Hamilton J Fantomos, The Phantom Limb.

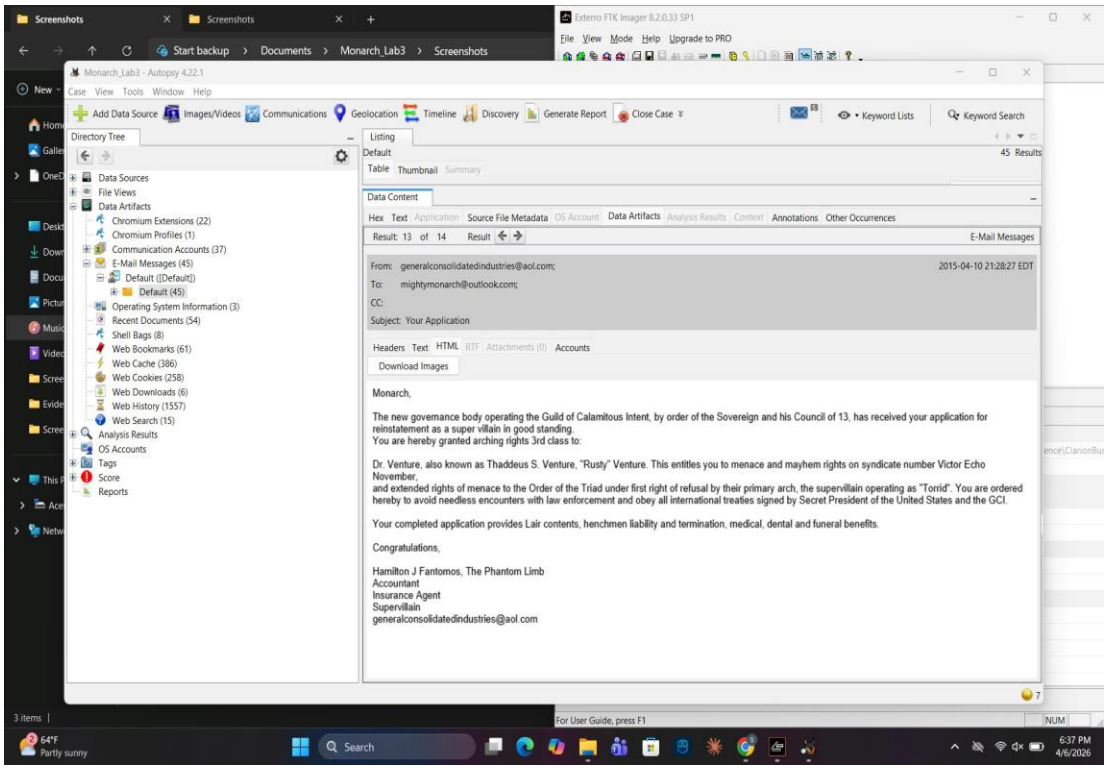

Figure 4: Email from Guild operative Hamilton J Fantomos

2) What is Malcom's email address? Did he send email to someone other than the Guild operative? Who?

Malcom's email address is **mightymonarch@outlook.com**. Yes, he sent emails to someone other than the Guild operative. He communicated with **Sheila at doctormrsthemonarch@outlook.com**, who appears to be his wife based on how she signs her emails ("Love, Sheila x0x0x").

I found this by looking at the E-Mail Messages in Autopsy. An email with subject "My new address" shows doctormrsthemonarch@outlook.com sending to mightymonarch@outlook.com. The message addresses him as "Malcolm" and is signed by Sheila.

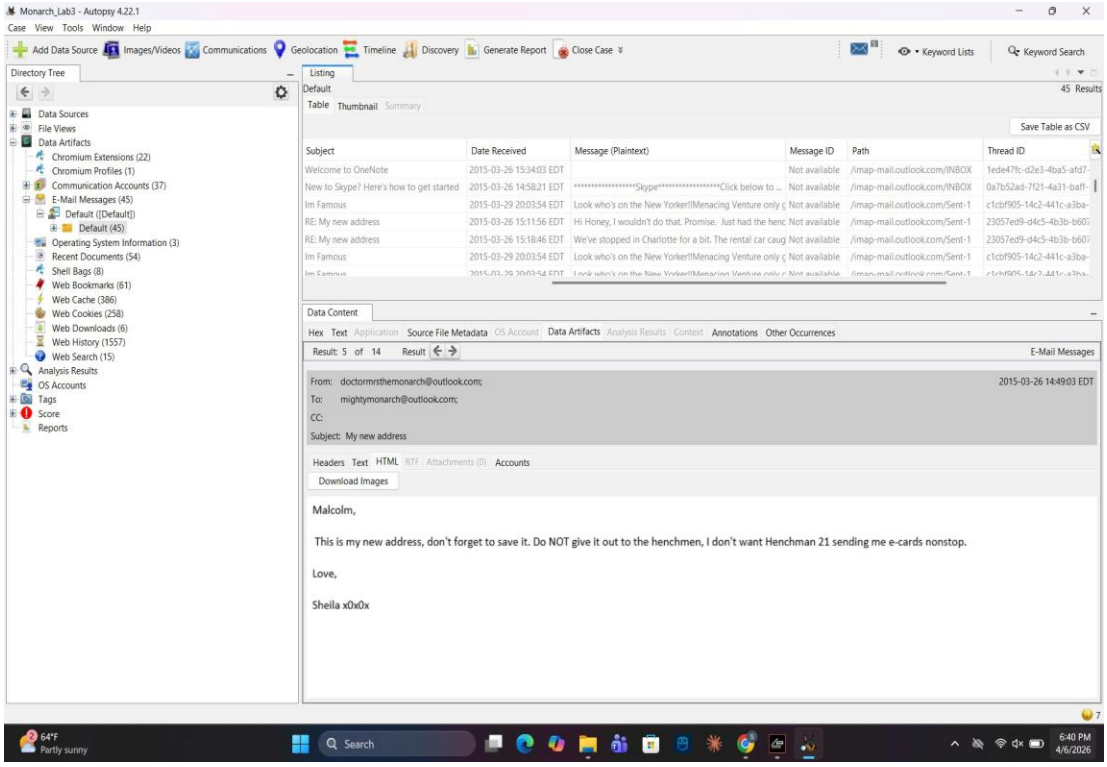

Figure 5: Email showing Malcom's address and communication with Sheila

3) What version OS was running on the computer in use?

The Monarch's computer was running **Windows 7 Enterprise with Service Pack 1 (Build 7601)**.

I found this by loading the SOFTWARE registry hive from the RegistryM folder into Registry Explorer. I navigated to Microsoft\Windows NT\CurrentVersion and found the ProductName value showing "Windows 7 Enterprise" and CSDVersion showing "Service Pack 1".

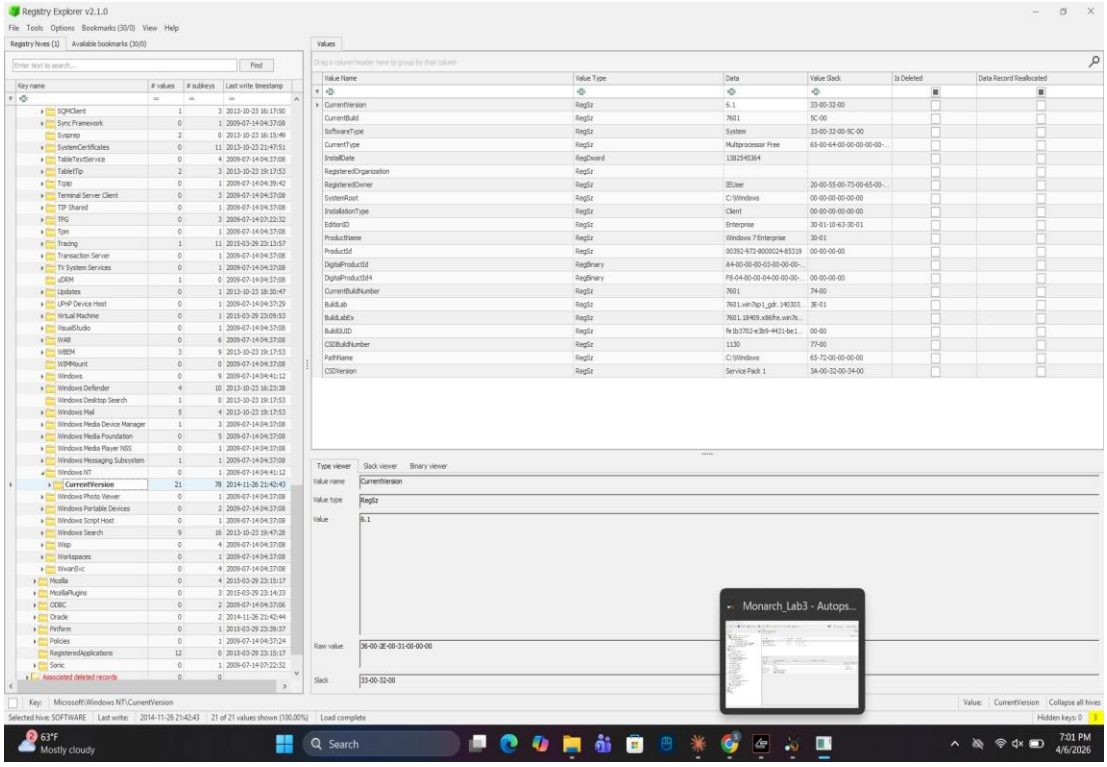

Figure 6: Registry showing Windows 7 Enterprise SP1

4) Did Malcom ever search for any terms that might mean he was going to cause any explosions?

Yes. Malcom searched for **"how to make fireworks"** on Bing using Microsoft Edge on 2015-03-30 at 00:22:41 EDT.

I found this by going to Data Artifacts > Web Search in Autopsy. The search term "how to make fireworks" was found in the WebCacheV01.dat file from the MonarchIMG2021.E01 image.

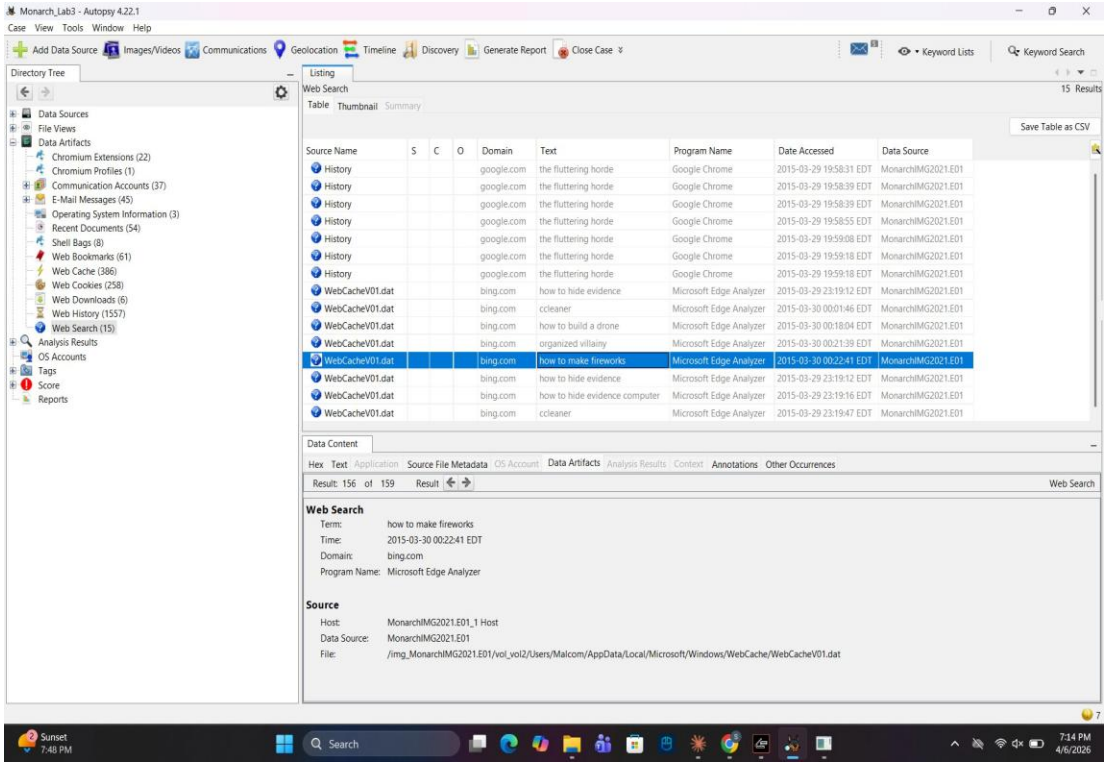

Figure 7: Web search for "how to make fireworks"

5) How about how to cover his tracks?

Yes. Malcom searched for multiple terms related to covering his tracks:
- **"how to hide evidence"** (2015-03-29 23:19:12 EDT)
- **"how to hide evidence computer"** (2015-03-29 23:19:16 EDT)
- **"ccleaner"** (2015-03-29 23:19:47 EDT)

CCleaner is a well-known tool that can be used to erase browsing history and other evidence from a computer. I found these searches in the Web Search artifacts in Autopsy from the WebCacheV01.dat file.

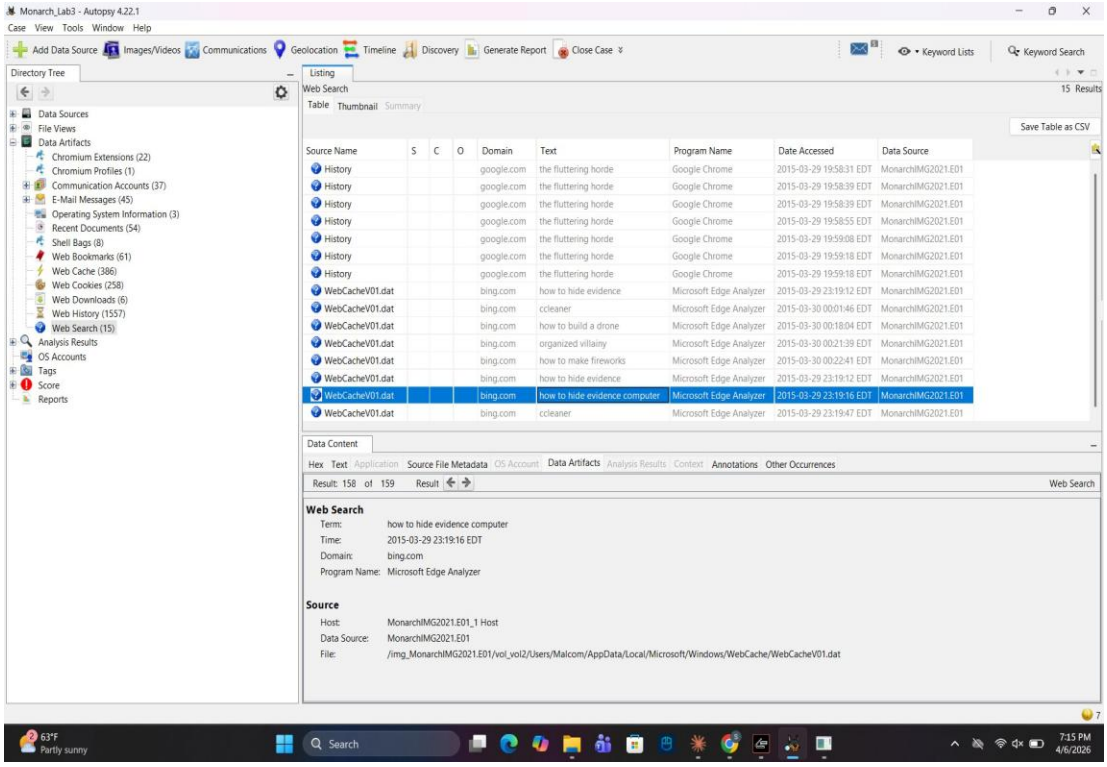

Figure 8: Web search for "how to hide evidence computer"

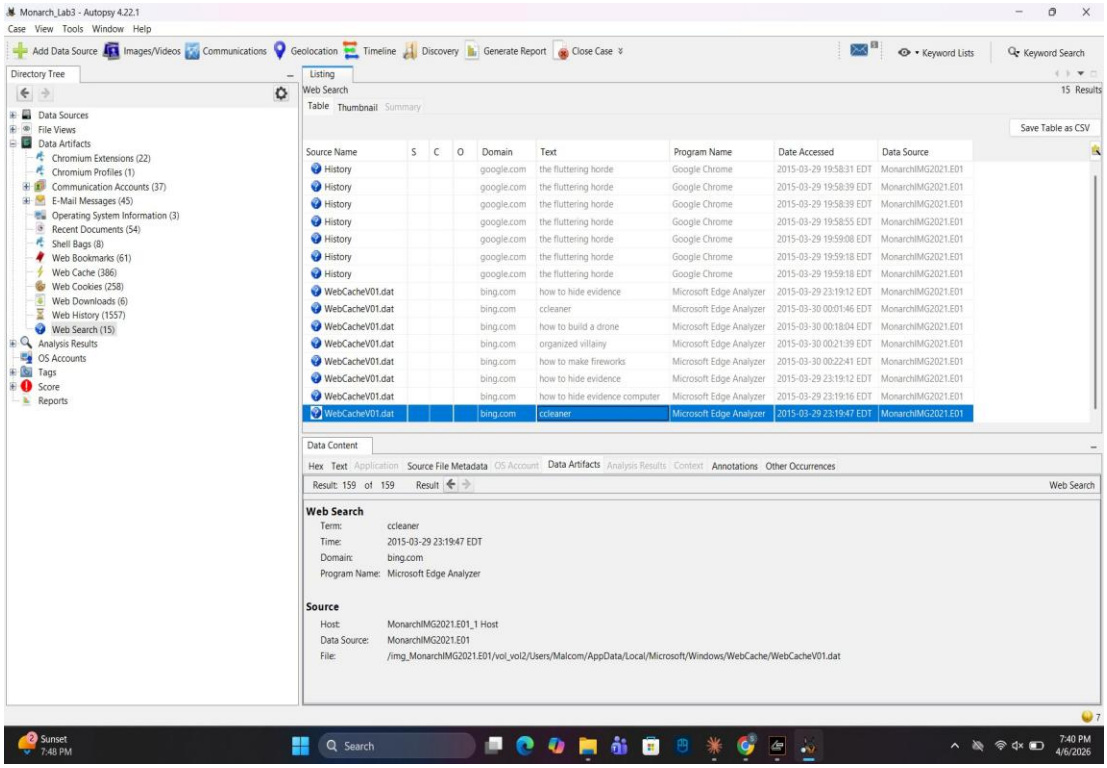

Figure 9: Web search for "ccleaner"

6) Explain in detail what he was doing, with what software and when.

Based on my analysis, here is a timeline of Malcom's activity:

March 29, 2015:
- 7:13-7:15 PM: Set up new computer, installed Google Chrome and Mozilla Thunderbird
- 7:58-7:59 PM: Used Google Chrome to search for "the fluttering horde" and downloaded images of The Monarch
- 8:03-8:06 PM: Sent emails to Sheila including one titled "Im Famous" about being on the New Yorker and Guild sanctioned
- 11:19 PM: Used Microsoft Edge/IE to search for "how to hide evidence" and "ccleaner"

March 30, 2015:
- 12:01 AM: Searched for "ccleaner" again
- 12:18 AM: Searched for "how to build a drone"
- 12:21 AM: Searched for "organized villainy"
- 12:22 AM: Searched for "how to make fireworks"

April 10, 2015:
- Received emails from Guild operative Hamilton J Fantomos regarding his application
- Sent angry replies to Phantom Limb

Software used: **Google Chrome, Microsoft Edge/Internet Explorer, Mozilla Thunderbird (email), and he downloaded CCleaner (ccsetup504.exe).**

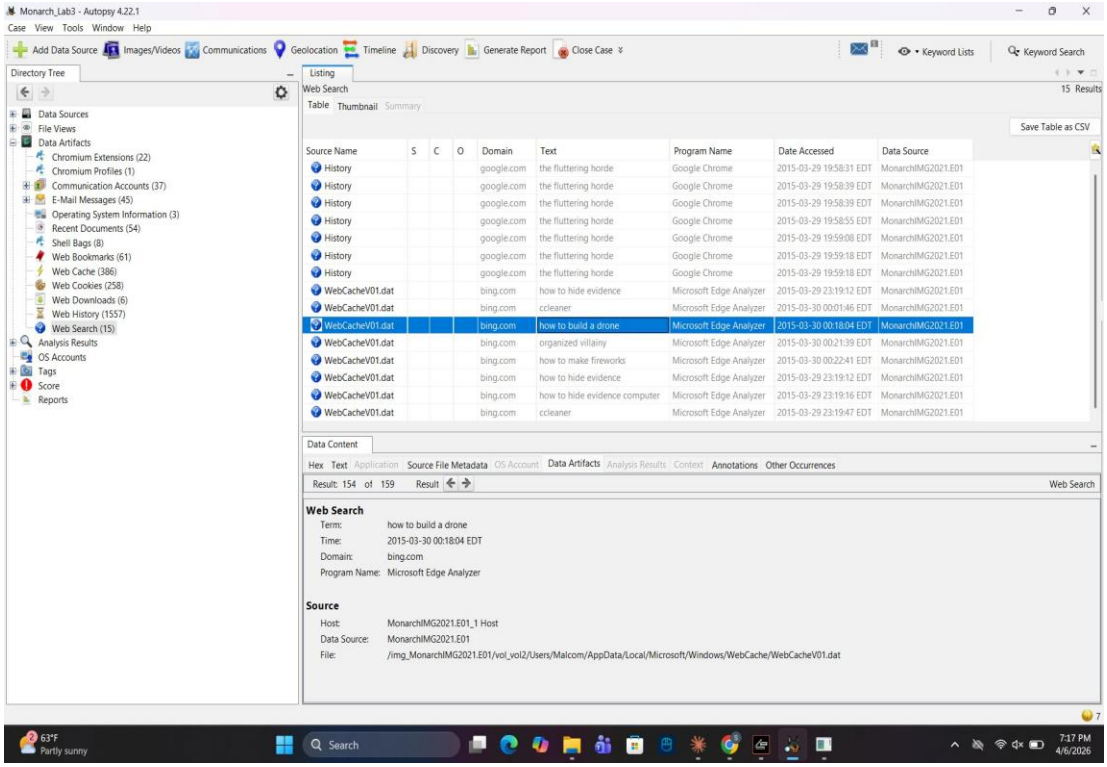

Figure 10: Web search for "how to build a drone"

7) Are there any photos that might have been downloaded? How do you know the photo was downloaded and not just moved to the folder where it was found? Is there a photo at this location that might tell us how he appears?

Yes, several photos were downloaded. **In the Downloads folder (Users\Malcom\Downloads), I found:**
- the_monarch_wants_you___again_by_petex-d2yvilz.jpg
- images.jpg
- venture-bros-monarch.jpg
- images (2).jpg

How I know they were downloaded and not moved: The Web Downloads artifact in Autopsy shows these files with their source URLs. For example, venture-bros-monarch.jpg was downloaded from venturebrosblog.com. Additionally, files downloaded from the internet in Windows get a Zone.Identifier Alternate Data Stream that marks them as coming from the Internet Zone - this proves they were downloaded, not just moved from another location on the computer.

Photo showing his appearance: Yes, the file venture-bros-monarch.jpg shows The Monarch's appearance - a character in a yellow and black monarch butterfly themed costume.

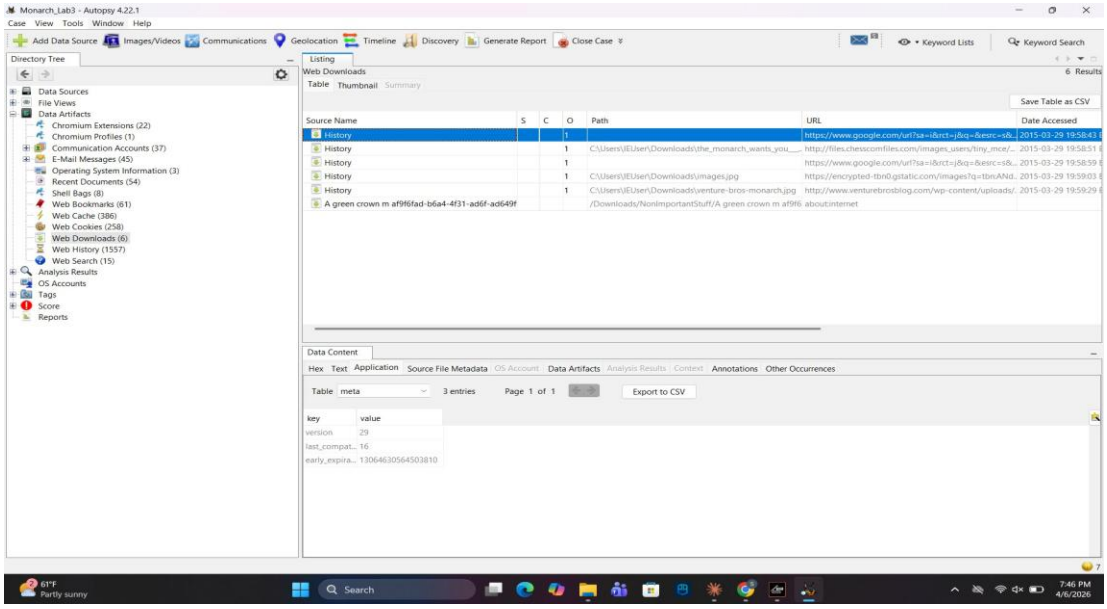

Figure 11: Web Downloads showing downloaded image files

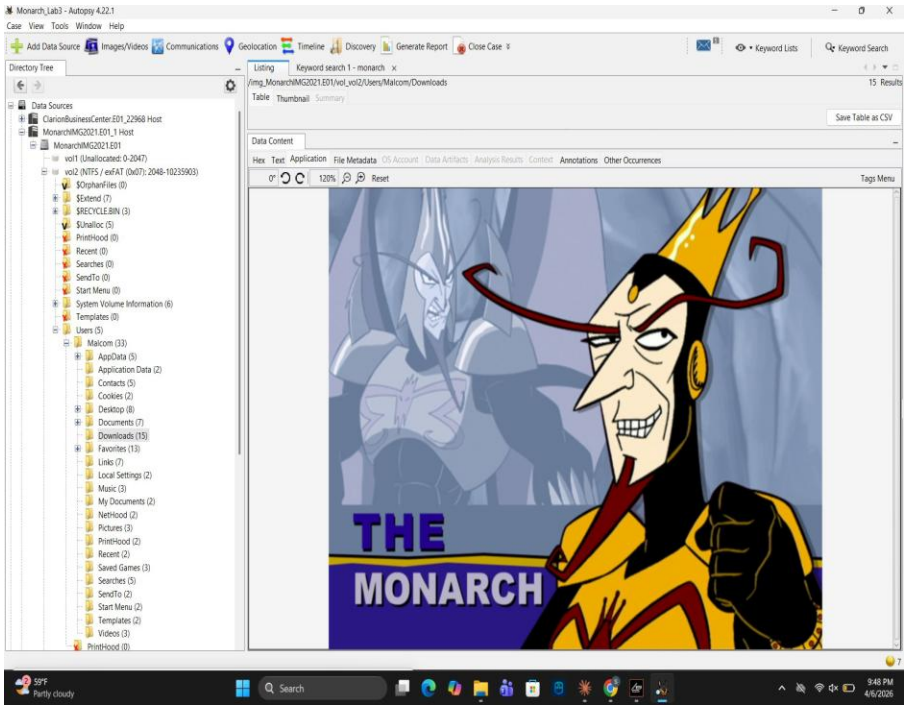

Figure 12: Image of The Monarch found in Downloads folder

8) Did Malcom do any research into forensics software? What specifically?

Yes, Malcom researched forensics software. Specifically:
- Searched for "ftk imager" on Google (2015-03-29 20:27:12)
- Visited AccessData.com to view the FTK Imager version 3.4.0 download page
- Visited magnetforensics.com reading about "How to uncover the covered tracks - hidden evidence using computer forensics tools"
- Visited Wikipedia article on "Anti-computer forensics"

I found this by searching for "ftk" and "forensic" in the keyword search in Autopsy. The results showed web history artifacts and cache files referencing these forensics tools and websites.

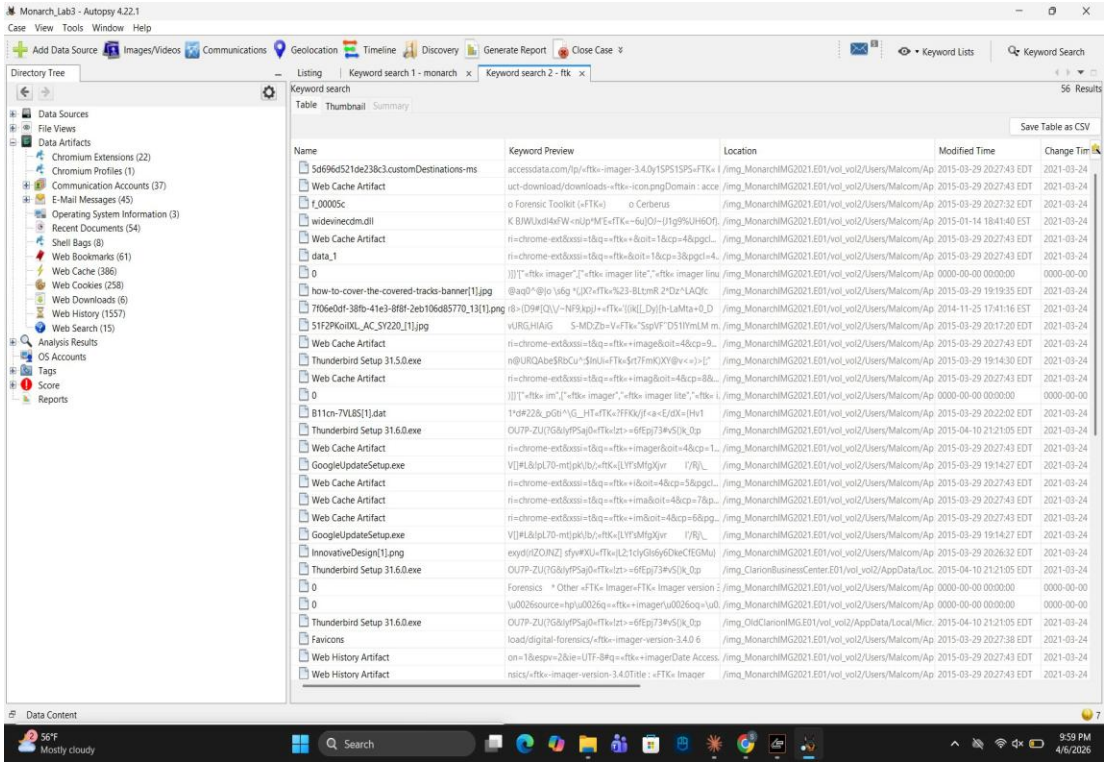

Figure 13: Keyword search results showing FTK Imager research

9) Is there any evidence he may have opened photos locally with an internet browser? Are those photos still available on the system?

Yes, there is evidence that Malcom opened local photos using an internet browser. **I found file:// URLs in the Web History showing:**
- file:///C:/Users/IEUser/Downloads/the_monarch...jpg
- file:///C:/Users/IEUser/Downloads/images.jpg
- file:///C:/Users/IEUser/Downloads/venture-br...jpg

The file:// protocol in a browser history indicates that local files were opened in the browser rather than web pages.

Are these photos still available? Yes, all three images are still present in the Users\Malcom\Downloads folder on the system. I was able to view them in Autopsy.

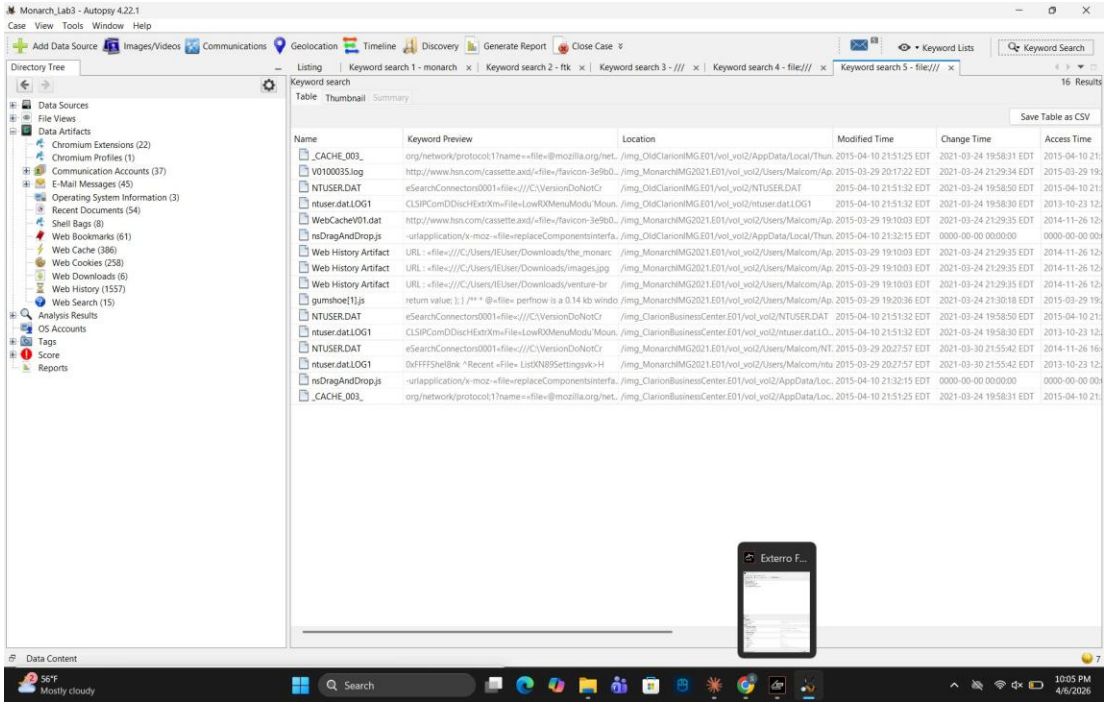

Figure 14: Keyword search showing file:/// URLs for local photos opened in browser

Conclusion

In this examination, I analyzed three forensic images related to the investigation of Malcom Monarch. All images were verified using FTK Imager with matching hash values confirming their integrity.

The examination revealed communication between Malcom (mightymonarch@outlook.com) and a Guild of Calamitous Intent operative named Hamilton J Fantomos (generalconsolidatedindustries@aol.com). Malcom also communicated with Sheila (doctormrsthemonarch@outlook.com).

The subject's computer was running Windows 7 Enterprise SP1. Web search analysis revealed searches for "how to make fireworks," multiple searches related to hiding evidence and CCleaner, and research into forensic tools including FTK Imager and anti-forensics techniques.

Downloaded images were found in the Downloads folder with evidence confirming they were downloaded from the internet. The subject also opened these local images using a web browser, and the files remain on the system.

Sam Suther
4/6/2026